



Your private AI companion for *motherhood*.

One calm conversation for trying to conceive, pregnancy, postpartum, and everything between — chat-first, with a hard clinical safety gate on every input and visible consent and privacy controls.

3

Surfaces — Android,
Web SPA, Admin
console

213

Backend tests
passing (18 files,
97% coverage)

20/20

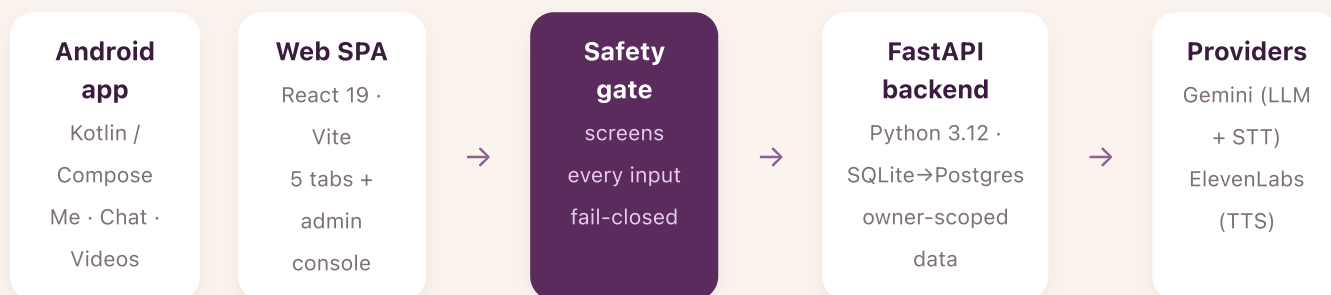
Safety evals on first
run, zero critical
failures

0

Third-party Android
dependencies

What Aira is

Aira is an AI maternal-wellness companion. A learner tells Aira where she is in her journey — trying to conceive, pregnant, or postpartum — and Aira shapes a proactive daily home, a gated AI conversation, week-by-week journey guidance, care reminders, and short own-hosted videos around that stage, in **English, Hindi, or Hinglish**. It offers wellness support, never diagnosis; anything that looks like a danger sign escalates to a human pathway instead of an AI answer.



Two clients, one backend. The web SPA is the reference implementation; Android is the lead client, verified on hardware, built with zero third-party dependencies (framework HTTP, JSON, VideoView, AlarmManager, RemoteViews throughout).

Non-negotiables

Safety gate on every input

The server-side gate screens **every** chat input before any reply exists — typed or spoken. It is fail-closed: if the gate errors, no AI reply may be generated. A red-flag input gets no AI answer at all — only the full-screen Urgent Help handoff with pathways to human care.

Learner data is owner-scoped

Every table is owned by the learner: exported on request, deletable, and merged when a guest signs in. Every new resource must wire into merge / erase / export before it ships. The aggregate metrics table has no learner column by construction.

Honest UI

No fake data anywhere. Trust labels reflect real gate results; star ratings are real aggregates; a video with no file shows an honest empty state rather than pretending to be playable.

Never used for advertising

Sensitive health data is not an ad product. AI personalisation is a consent with real effect: switched off, no memories enter prompts and no extraction runs.

Features — safety, conversation, and the proactive home

SAFETY & TRUST

- **Clinical safety gate** screening every turn (Gemini safety model), fail-closed with an honest "safety check unavailable" state
- **Urgent Help takeover** — full-screen escalation, call pathways, emergency profile, "I'm safe for now"
- **Per-message safety chip** reflecting the real gate decision
- **Safety evals suite** — 20/20 on first run: typos, Hinglish, Hindi negation, third-party reports, idioms; p95 gate latency 1.7 s
- **Admin safety review queue** — filterable, paginated, append-only gate record, every action logged
- **AI disclaimer** served by the backend and surfaced beside AI replies

CHAT & AI

- **Gated AI turns** with empathy-first replies and typed content cards
- **SSE streaming** on web; blocking POST /respond on Android
- **Three languages** — English, Hindi, Hinglish — served from one backend catalog (/config), including Hinglish cards
- **Voice loop** — speech → Gemini STT → the same gated spine → ElevenLabs TTS reply, LRU-cached; silence and STT failures degrade honestly
- **Memory extraction** with a review surface — the learner sees and controls what Aira remembers
- **Next-best-action card** based on the learner's actual day
- **Typing indicator, multi-line input, quick chips**, onboarding that happens inside the conversation

ME — THE PROACTIVE HOME

- **Time-aware hero** — Week · Day, baby size (emoji + ~cm), days/weeks to go, journey progress bar, next-milestone pill
- **Mood check-in** — six moods with per-mood colour and icon, notes, 7-day strip, monthly breakdown, trend escalation, postpartum AM/PM slots
- **Today's care** — water droplets with an editable goal, tick/untick with undo, per-event reminder times
- **Daily tip** — rotating, language-aware with English fallback, thumbs feedback
- **Learned nudge ranking** — server-ranked focus nudges that learn from taps; a nudge never duplicates a control already on screen
- **Streaks, Sunday recap, week-flip celebration**, name prompt, pull-to-refresh, offline banner

IN-CHAT TOOLS (all on real endpoints)

- Check-in
- Reminders
- Care Vault uploads
- Reset (2 calm minutes)
- Symptom tracking
- Medicines
- Visit copilot + calendar
- Care plan
- Memory review
- Companion mode
- Emergency profile
- Calm-match game

The "+" tray in chat opens seven everyday tools; the rest are reachable from Me and Settings.

Features — journey, videos, care, privacy, platform

JOURNEY

- **Week-by-week guide** — editorial sections (your body, prepare for your visit), size comparisons, browsing weeks without mutating the care context
- **Milestone path timeline** — a Duolingo-style path with "you are here"
- **Change stage anytime** — two-tap stage picker, no re-onboarding
- **Journey CMS** — admin edits reach learners immediately

VIDEOS

- **Own-hosted short videos** streamed with HTTP Range support (seekable), portrait + landscape variants
- **Stage-aware catalog** with search, category chips, duration badges
- **Likes and 1–5 star ratings** — real aggregates only
- **In-app player** — completion marks watched and drives unwatched-first rotation; transcript toggle for sound-off and screen-reader use

CARE

- **Reminders** with progress, undo, editable targets, and per-event notification times (bell editor)
- **Medicines** — doses, schedules, taken-today state
- **Visit copilot** — appointments, questions and notes, add-to-calendar
- **Care Vault** — prescription and report uploads
- **Wellness reports** — monthly done-day summaries

PRIVACY & ACCOUNTS

- **AI-personalisation consent with real effect** — off means no memories in prompts and no extraction
- **Export and erase** — learner data leaves or disappears on request
- **Guest-first** — device tokens; sign-in merges the guest history
- **Owner-scoped tables** throughout; aggregate metrics carry no learner column

PLATFORM & SYSTEM SURFACES

- **Dark mode** — a switchable palette where every colour resolves through a state getter; System / Light / Dark picker
- **Opt-in daily notification** (3 PM, AlarmManager) with enable-time preview
- **Home-screen widget** fed from a local cache, with a quick action
- **Share card** — image render of the week
- **Admin console** — owner/support/content roles, PBKDF2 passwords, HMAC admin tokens disjoint from learner tokens
- **Deploy hardening** — Dockerfile, smoke test, rate limiting, history sanitisation

QUALITY

- **213 backend tests** across 18 files; ~97% coverage
- **Live-verified loops** — chat, voice, admin review; real bugs found live were fixed with regression tests
- **Device-verified** Android on hardware; zero third-party dependencies

Design system — the "Dawn" direction, shipped

Aira's visual language pairs an editorial serif (Fraunces-class) with a calm plum/lilac/sage palette. The **Dawn redesign** — proposed in an HTML before/after kit and then implemented move by move in the Compose app — made six moves:

1 • Twilight → Dawn

The dark hero becomes a light sunrise gradient (peach → lilac) with plum ink; the dark theme keeps its dark hero for 3 AM feeds.

2 • Shadows replace borders

Cards, bubbles, and tiles drop 1px outlines for soft plum-tinted shadows and larger radii.

3 • A warm second accent

Rose joins plum on eyebrows, tool tiles, and category chips; sage keeps meaning "done / safe".

4 • Frosted nav

The floating nav pill goes translucent white with a solid plum active tab.

5 • Editorial serif

Italics for greetings and emphasis words; the welcome headline grows. Companion voice, not system chrome.

6 • Rounder controls

Buttons, options, segments, mood chips, and check buttons go full-pill or circle; the primary button gains a plum gradient.

The design team works from docs/design/index.html (all 13 screens as living HTML, token-driven) and docs/design/redesign.html (before/after pairs) — editing one CSS token block re-skins every screen.

Current status & launch gates

Complete

All build phases (P0–P11), the 55-point world-class home review, the Dawn redesign, admin + deploy hardening, and the voice loop — with the full product loop, voice loop, and admin review loop verified live.

Human-gated before launch

- **Clinician review** of the safety taxonomy, seeded journey content, and videos
- **Privacy / jurisdiction legal review**
- **Production infrastructure** — Postgres, Redis, object storage

Tracked with lead times in TODO.md. These are deliberate human gates, not engineering gaps.

COMPONENT	STATE
Backend (FastAPI)	213 tests green; Docker + smoke test
Android (Compose)	Lead client, device-verified, Dawn shipped
Web (React 19)	Reference client + admin console
Design kit (HTML)	13 screens, token-driven, in sync